Altium Designer 24

Pocket reference for PCB designers

Shortcuts, panels, layers, Footprints, Polygon Pours, DRC, Gerber, NC Drill and PCBWay preflight.

Explanations in English; Altium menu labels, abbreviations and commands remain exactly as in the application. For Altium Designer 24.x, two-layer boards and everyday PCB work.

Start here — 12 essential actions

Command / shortcut	What it does / when to use it
2 / 3	PCB 2D / 3D view.
Shift + S	Show only the active layer (Single Layer Mode); repeat to cycle modes.
Q	Toggle mm ↔ mil in PCB Editor.
F11	Show / hide Properties.
Ctrl + M	Measure the distance between two points.
Ctrl + W	Start Interactive Routing.
Ctrl + PgDn	Open View Configuration: Layers & Colors.
V, F	Fit all objects in the current document.
D, R	Open PCB Rules and Constraints Editor using the menu.
T, D, R	Run the batch Design Rule Check using the menu.
T, G, M	Open Polygon Pour Manager.
P, S	Place a String on the current layer.

TIP: Ctrl + W means pressing keys together. Comma-separated menu accelerators (such as T, G, M) are pressed in sequence and can vary by editor or version. Use the English keyboard layout.

1. Display, navigation and object selection

Command / shortcut	What it does / when to use it
2 / 3 / 1	2D Layout / 3D Layout / Board Planning Mode.
Ctrl + F	Flip Board View (view only; does not move components).
Shift + S	Single Layer Mode changes visibility, NOT the active layer.
Num + / Num −	Go to the next / previous enabled layer.

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Num *	Go to the next signal layer.
Ctrl + Shift + Wheel	Cycle active layers with the mouse wheel (default mapping).
Shift + C	Clear active filters / masking.
F11 / F12	Properties / PCB Filter.
Double-click	Open Properties for the object under the pointer.
Shift + F	Find Similar Objects — select related objects in bulk.
Esc / Right-click	Cancel current command; sometimes press twice.

TIP: If Top Layer seems missing, inspect the layer tab at the bottom of PCB, Single Layer Mode, and View Configuration (Panels → View Configuration).

2. PCB layers: avoid confusion

Command / shortcut	What it does / when to use it
Top Layer / Bottom Layer	Top / bottom copper (typically red / blue, depending on display settings).
Top Overlay / Bottom Overlay	Silkscreen: component labels and outlines.
Top Solder / Bottom Solder	Solder mask openings around pads.
Top Paste / Bottom Paste	Stencil openings for solder paste; NOT silkscreen.
Mechanical Layer	Mechanical layers: purpose depends on the project.
Keep-Out Layer	Restricted areas; do not confuse with the actual board outline.
Layer Stack Manager	Configure the stackup, copper layers, dielectric materials and thicknesses.

TIP: Names such as GND_TOP and GND_BOT do not determine the actual layer. Check each Polygon Pour's Layer property.

3. Components, Footprint and Pad

Command / shortcut	What it does / when to use it
Click → Properties	Select a Pad (refine selection if necessary), then inspect X-Size, Y-Size, Hole Size, Net and Layer.
Space / Shift + Space	Rotate a moving object counterclockwise / clockwise.
L during Move	Flip the moving component to the opposite side of the PCB.
M, M	Move Object.
Q	Toggle mm / mil in PCB.
Ctrl + Q in Properties	Toggle measurement units in the Properties panel / dialog.
0.12mm	Always type an explicit unit suffix if the current display is in mil.
Tools → Update PCB With	In .PcbLib: push the current Footprint changes into

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Current Footprint	PCB.
Tools → Update From PCB Libraries	In PCB Editor: update from libraries; carefully review proposed changes.

Pad Stack: three different settings

Command / shortcut	What it does / when to use it
COPPER X-Size / Y-Size	Physical copper-pad width and height.
PASTE Expansion	Stencil aperture size for solder-paste printing.
SOLDER Expansion	Solder-mask aperture expansion around a copper pad.
Top / Bottom	Through-Hole Pad can have distinct mask settings for each side.
Hole Size / Plated	Drill diameter / plated-through-hole option.

TIP: In our case, 0.12 mm = 4.724 mil. Do not confuse 0.12 mil (= 0.003048 mm) with 0.12 mm. Increasing Mask Expansion can trigger new Minimum Solder Mask Sliver violations.

4. Placement and Interactive Routing

Command / shortcut	What it does / when to use it
P, T / Ctrl + W	Start Interactive Routing.
Shift + Space	During routing: cycle corner styles.
Space	During placement/routing: change corner direction or orientation, depending on context.
Shift + R	During routing: cycle obstacle-avoidance modes (Push, Walkaround, etc.).
Num + / Num − during Routing	During Routing: switch to an allowed signal layer and insert a Via.
Tab	During an interactive command: open its available settings.
Backspace	During routing: remove the most recently placed segment.
P → Via	Place a Via manually from the Place menu, when needed.

TIP: Shift + S does not switch Top/Bottom. Select the target layer first, then enable Single Layer Mode.

5. Polygon Pours — ground and power

Command / shortcut	What it does / when to use it
Place → Polygon Pour	Create a polygon outline on the chosen layer.
Tools → Polygon Pours →	List polygons and inspect Net, Layer, Shelved, Pour

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Polygon Manager	Order, etc.
T, G, M	Full menu path; letter accelerators can differ.
Shelved	Temporarily remove a polygon from poured copper; restore with Restore Shelved Polygons.
Repour All	Recalculate all pours after a board change.
Net = GND	Assign the electrical net (the polygon name alone has no effect).
Layer = Top Layer	Verify the actual layer for GND_TOP, regardless of its name.
Remove Dead Copper	Remove disconnected copper islands when enabled.

When a polygon is invisible

1. In Polygon Manager: does the polygon exist? Correct Layer? Shelved unchecked? 2. Repour All. 3. Is the correct layer visible? 4. Is Polygon visibility enabled in View Configuration? 5. Check the GND net and whether the polygon lies within the board.

6. Design Rules and DRC

Command / shortcut	What it does / when to use it
Design → Rules / D, R	Open PCB Rules and Constraints Editor.
Tools → Design Rule Check / T, D, R	Run batch DRC on the PCB.
Messages / PCB Rules and Violations	Review errors, affected objects and measured distances.
Clearance Constraint	Minimum spacing between copper objects.
Minimum Solder Mask Sliver	Minimum solder-mask web between neighboring openings.
Silk To Solder Mask Clearance	Silkscreen overlaps or approaches an exposed pad / mask opening.
Silk To Silk Clearance	One silkscreen element overlaps another.
Short-Circuit / Un-Routed Net	Electrical short / unrouted electrical connection.

Real-world example: AHDS RGB2

• The Pad had SOLDER Expansion = 0 mm, and CAMtastic reported Annular Ring (Pad → Mask).

• Raising this to 0.12 mm reduced CAM warnings but triggered Minimum Solder Mask Sliver violations on the PCB when the rule was 0.254 mm.

• For diagnosis the rule was set to 0.16 mm. The actual allowed value must be confirmed for the specific PCBWay order.

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• Updating the library restored silkscreen elements that overlapped pads; these were corrected without modifying copper.

TIP: DRC = 0 means no violations under the enabled rules. It does NOT guarantee manufacturability. Never relax rules only to force a zero.

7. Gerber, NC Drill, CAMtastic and ZIP

Command / shortcut	What it does / when to use it
File → Fabrication Outputs → Gerber Files	Generate Gerber from PCB Editor; the output path may depend on Output Job.
File → Fabrication Outputs → NC Drill Files	Generate NC Drill hole coordinates.
CAMtastic → File → Import → Quick Load	Import a fresh manufacturing set into a new CAM Document.
CAMtastic → Analysis → PCB Design Check/Fix	Analyze imported fabrication layers.
Project Outputs for ...	Common destination folder when using standard output generation.
GTL / GBL	Top Copper / Bottom Copper.
GTS / GBS	Top Solder Mask / Bottom Solder Mask.
GTO / GBO	Top Overlay / Bottom Overlay (silkscreen).
GM / GM1 / GKO	Board outline — file extension depends on output mapping; check actual geometry.
TXT / DRL	Usually NC Drill; verify file contents and generation date.
GTP / GBP	Paste Stencil Layers — normally unnecessary for bare-PCB fabrication.
DRR / REP / APR	Reports / aperture data, normally not included in a manufacturing ZIP.

Minimum package for a two-layer PCB

GTL + GBL + GTS + GBS + GTO + board outline (GM/GM1/GKO) + NC Drill (TXT/DRL). Include GBO if there is bottom silkscreen. Include separate PTH/NPTH drill files when generated. Check exact roles rather than relying on extensions alone.

Final checklist before PCBWay

1. Ctrl + S → Repour All → PCB DRC. 2. Regenerate Gerber. 3. Regenerate NC Drill. 4. New CAM Document + import fresh files. 5. Check layers, outline, drill holes, masks and silkscreen. 6. Verify board specifications with PCBWay before ordering.

TIP: Existing CAM documents can retain previously imported data. Updating a file on disk does not necessarily reload it in CAMtastic.

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8. Quick troubleshooting — symptom → action

Command / shortcut	What it does / when to use it
Polygon missing	Polygon Manager: Layer / Shelved / Net → Repour All → View Configuration.
Cannot select Pad	Zoom in, refine the object under the cursor, open Properties; inspect Selection Filter.
0.003 mm shown instead of 0.12 mm	mm / mil mix-up; enter 0.12mm explicitly and verify conversion.
Changed .PcbLib, PCB unchanged	Update PCB With Current Footprint or Update From PCB Libraries; inspect ECO.
DRC reports Silk… again	Inspect Footprint Overlay versus mask openings; move / trim the silkscreen.
CAMtastic shows old warnings	Regenerate Gerber + NC Drill, make a new CAM Document and import afresh.
Text is not visible on PCB	Check Layer = Top Overlay and its visibility.
CAMtastic: Pad → Mask	Check actual Mask Expansion and CAM threshold; avoid increasing it without checking Mask Sliver.

9. Personal checklist before your first project

Follow your familiar OrCAD / PADS / Eagle workflow: schematic → footprint/library audit → rules → placement → routing → polygon pour → DRC → manufacturing outputs. Altium changes the interface, not the underlying engineering discipline.

Official references for command verification

Altium Designer — PCB Editor & PCB Library Editor Shortcuts:
https://www.altium.com/documentation/altium-designer/shortcut-keys/pcb-editors

Altium Designer — Shortcut Keys and PCB sequences:
https://www.altium.com/documentation/altium-designer/shortcut-keys

Note: Menu positions and certain commands can vary by build, workspace settings or document type. If a command is missing, use Search or check its editor context.

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